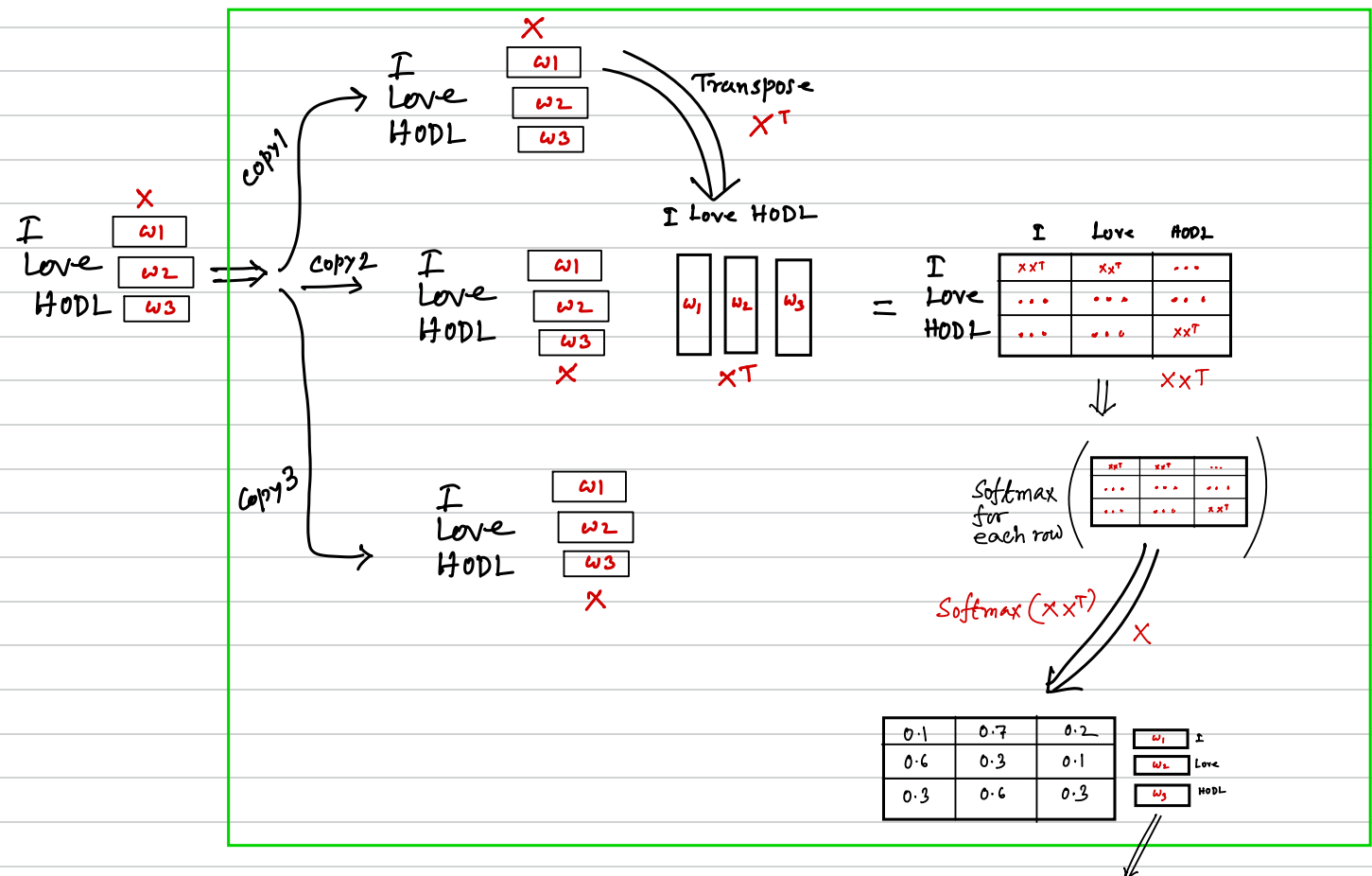
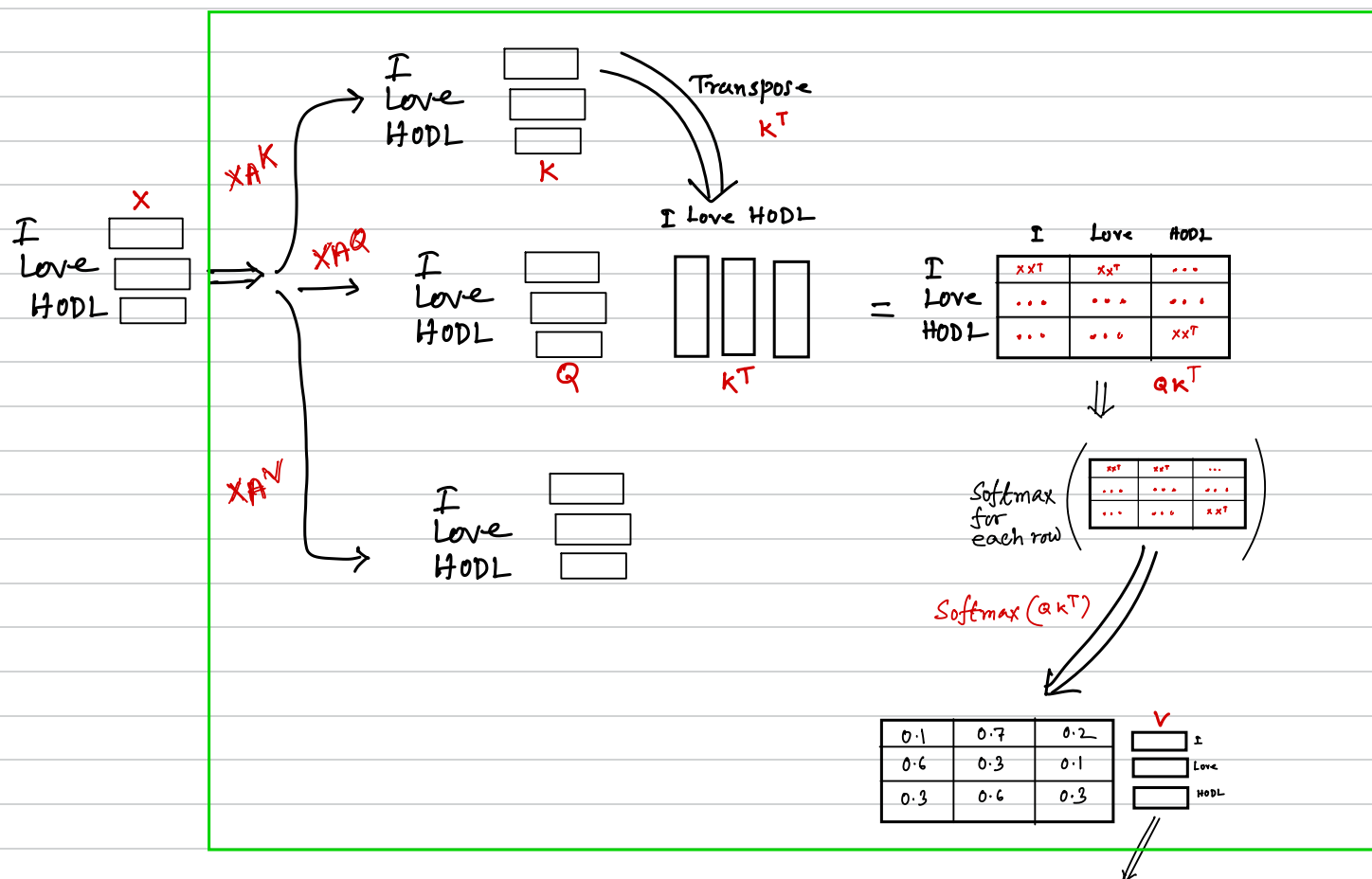


A Self Attention Layer



$$\text{Softmax}(XW)X = \begin{matrix} & \begin{matrix} I & Love & HODL \end{matrix} \\ \begin{matrix} I \\ Love \\ HODL \end{matrix} & \begin{bmatrix} 0.1w_1 + 0.7w_2 + 0.2w_3 \\ 0.6w_1 + 0.3w_2 + 0.1w_3 \\ 0.3w_1 + 0.6w_2 + 0.3w_3 \end{bmatrix} \end{matrix}$$

The Tunable Self-Attention Layer: (K, Q, V)



$$\text{activation}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_K}}\right)V$$

I	$0.1w_1 + 0.7w_2 + 0.2w_3$	$\hat{w}_1$
Love	$0.6w_1 + 0.3w_2 + 0.1w_3$	$\hat{w}_2$
HODL	$0.3w_1 + 0.6w_2 + 0.3w_3$	$\hat{w}_3$